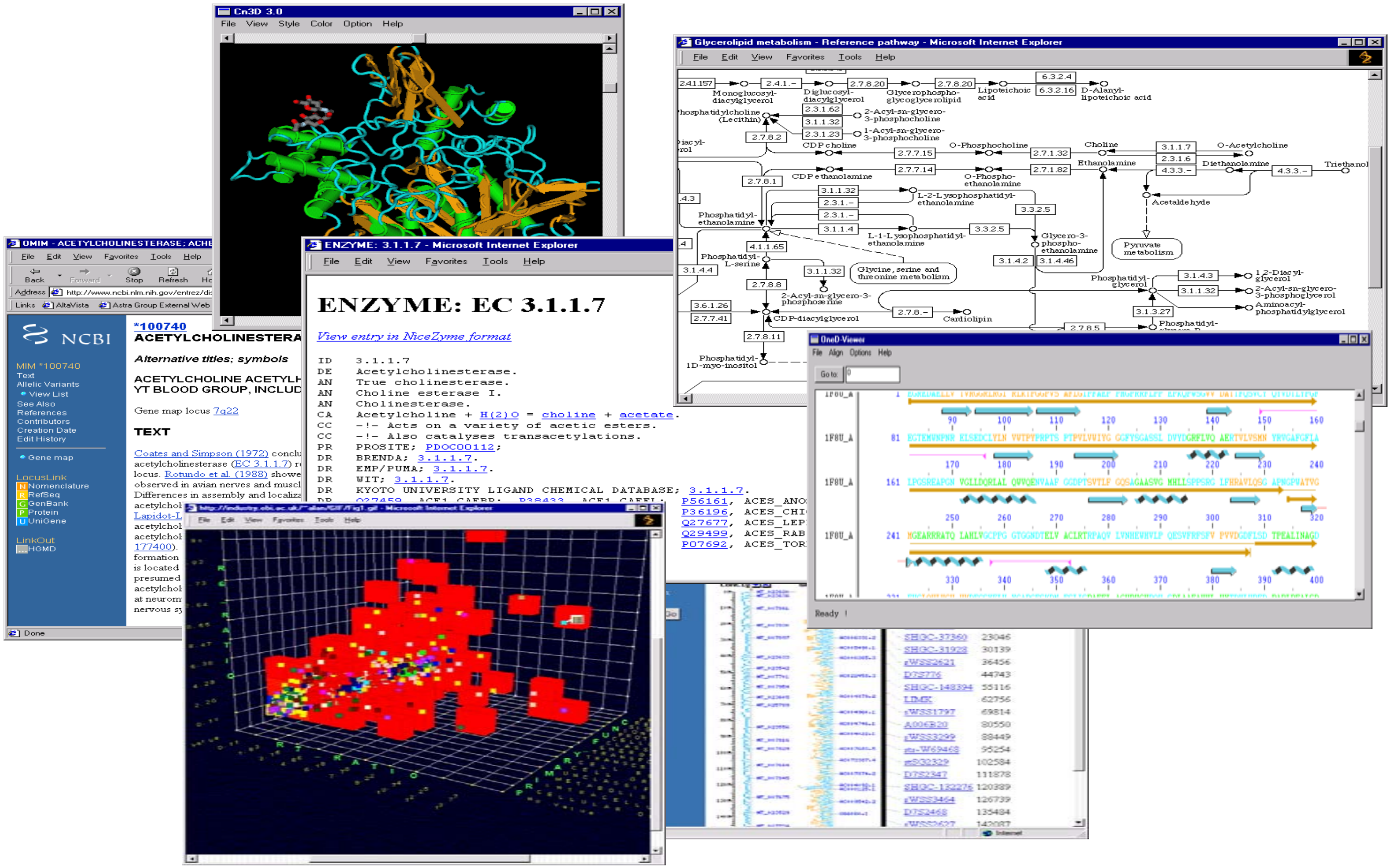
# Managing and Integrating Discovery Data with Semantic Technologies

Susie Stephens
Server Technology

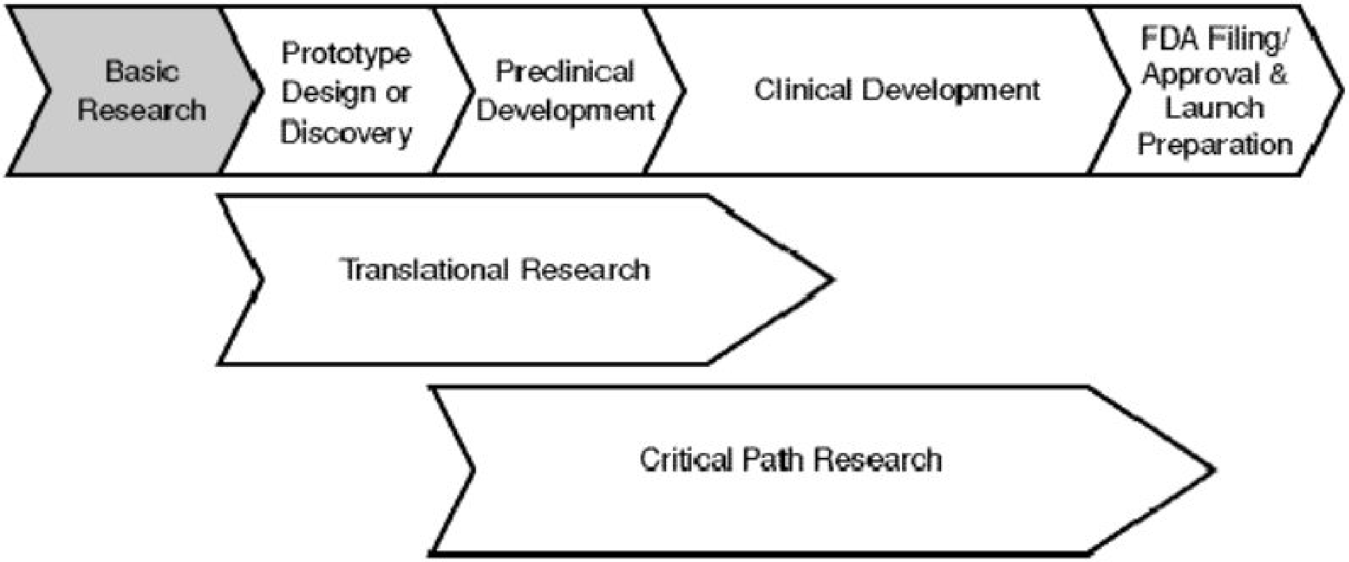
# Presentation Outline

- Data Integration in the Life Sciences
- Characterizing the Semantic Web
- Oracle RDF Data Model
- Customer Use Cases
- W3C's Work in Health Care and Life Sciences

# The Complexity of Discovery Data



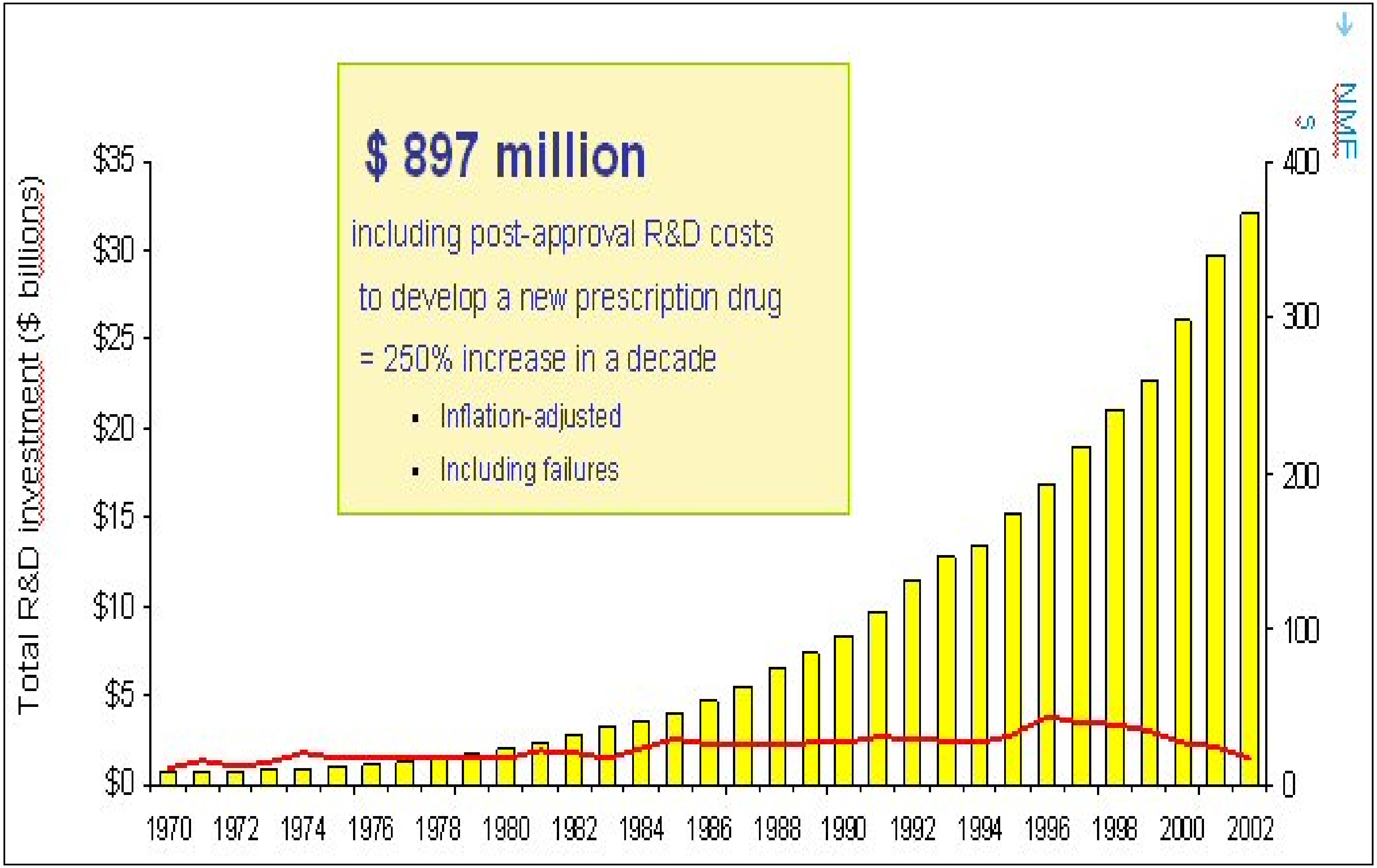
# Critical Path Initiative




"In response to the widening gap between basic biomedical knowledge and clinical application, governments and the academic community have undertaken a range of initiatives. After decades of investment in basic biomedical research, the focus is widening to include *translational research* — multidisciplinary scientific efforts directed at "accelerating therapy development" (i.e., moving basic discoveries into the clinic more efficiently)."

Source: Innovation or Stagnation, FDA Report, March 2004

# Pharma Productivity






Source: PhRMA & FDA 2003

# Characterizing the Semantic Web

- Semantic Web is an integration technology
- Standardization is the traditional approach to data integration
- Semantic Web has a different approach
  - Less focus on *a priori* standardization
  - Emphasis on how to say things, not what to say
- Integration with shared and accessible semantics
- Greater level of future proofing and reuse
- Semantic Web aims at achieving things *ad hoc*

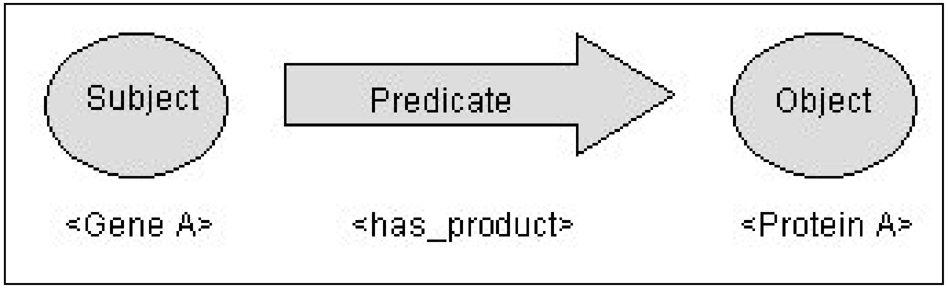
**ser•en•dip•i•ty** |ˌserənˈdipitē|

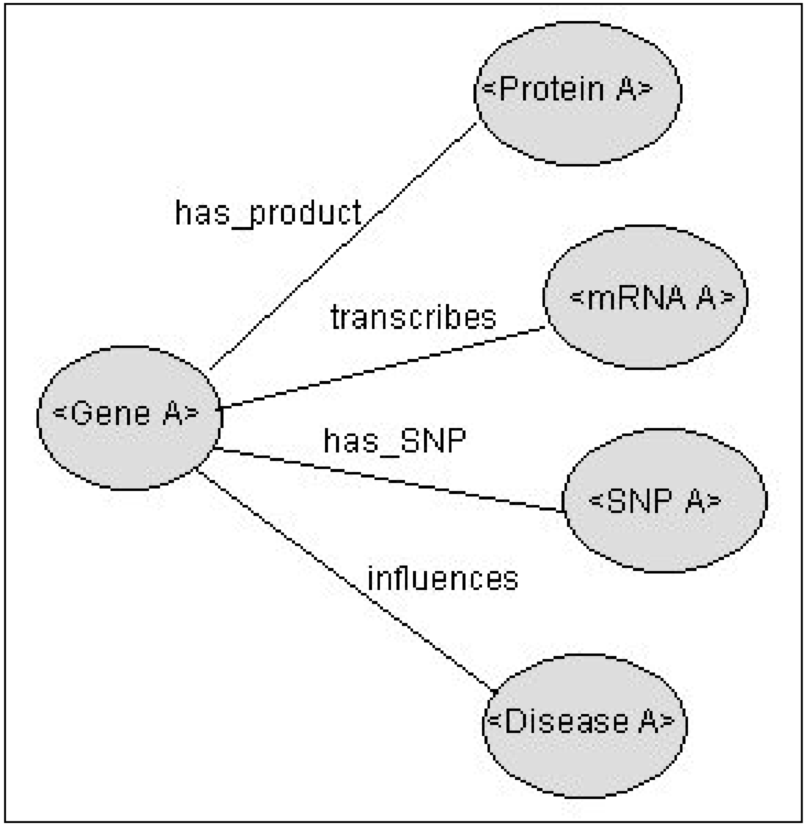
noun

the occurrence and development of events by chance in a happy or beneficial way: *a fortunate stroke of serendipity* | *a series of small serendipities*

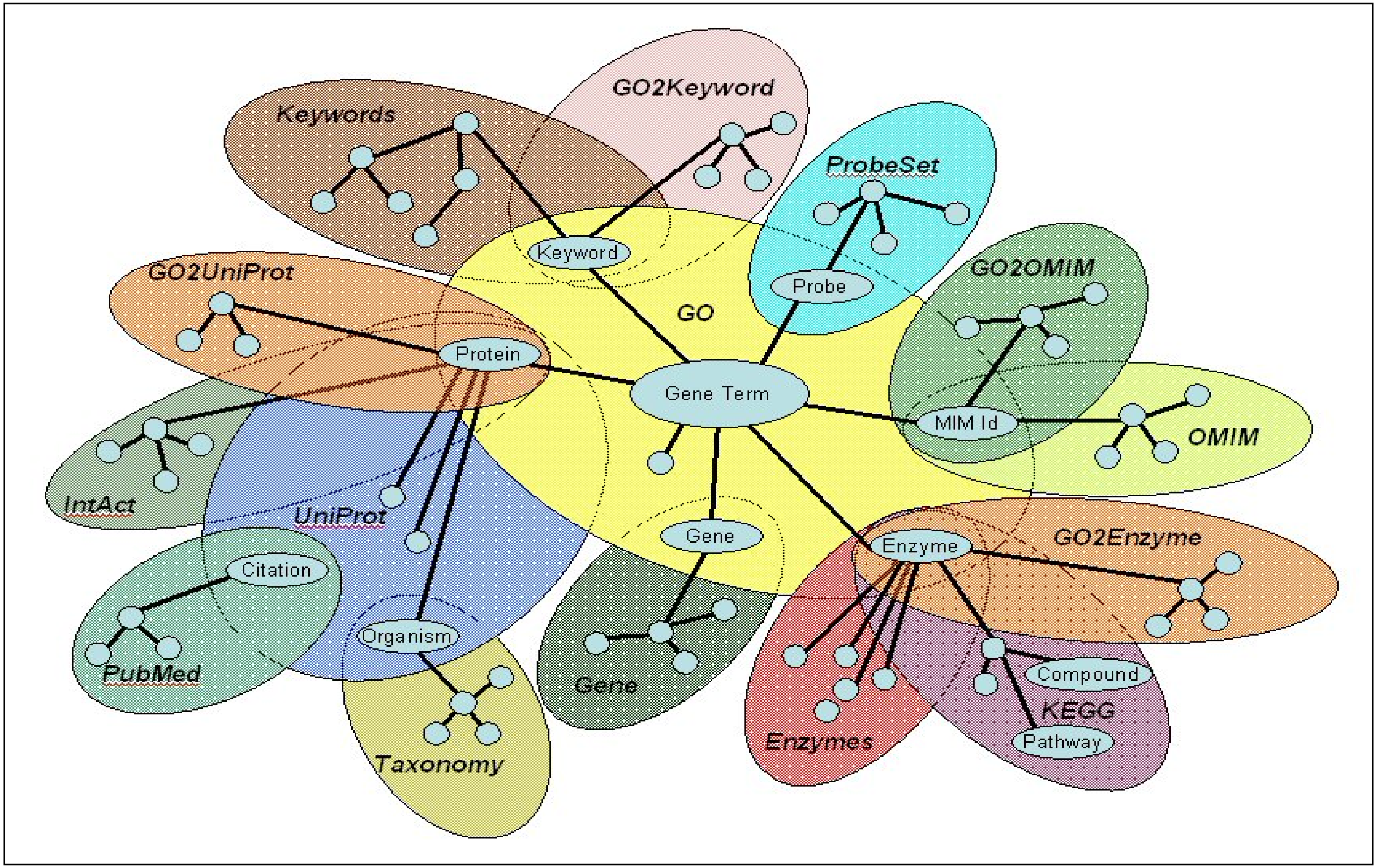
Source: Oxford American Dictionary

# Resource Description Framework (RDF)



# RDF-based Data Integration






Source: J Web Semantics

# Web Ontology Language (OWL)

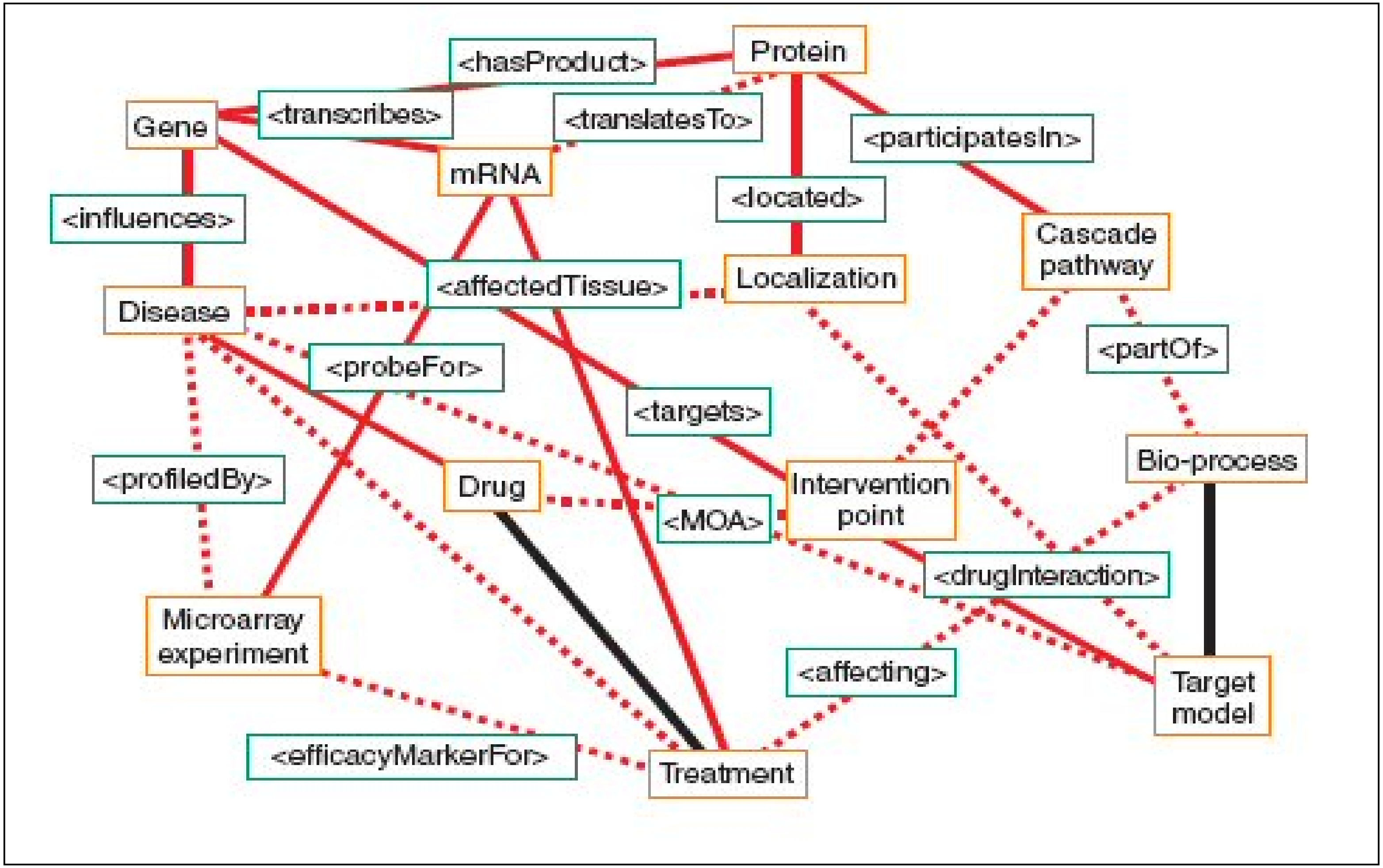
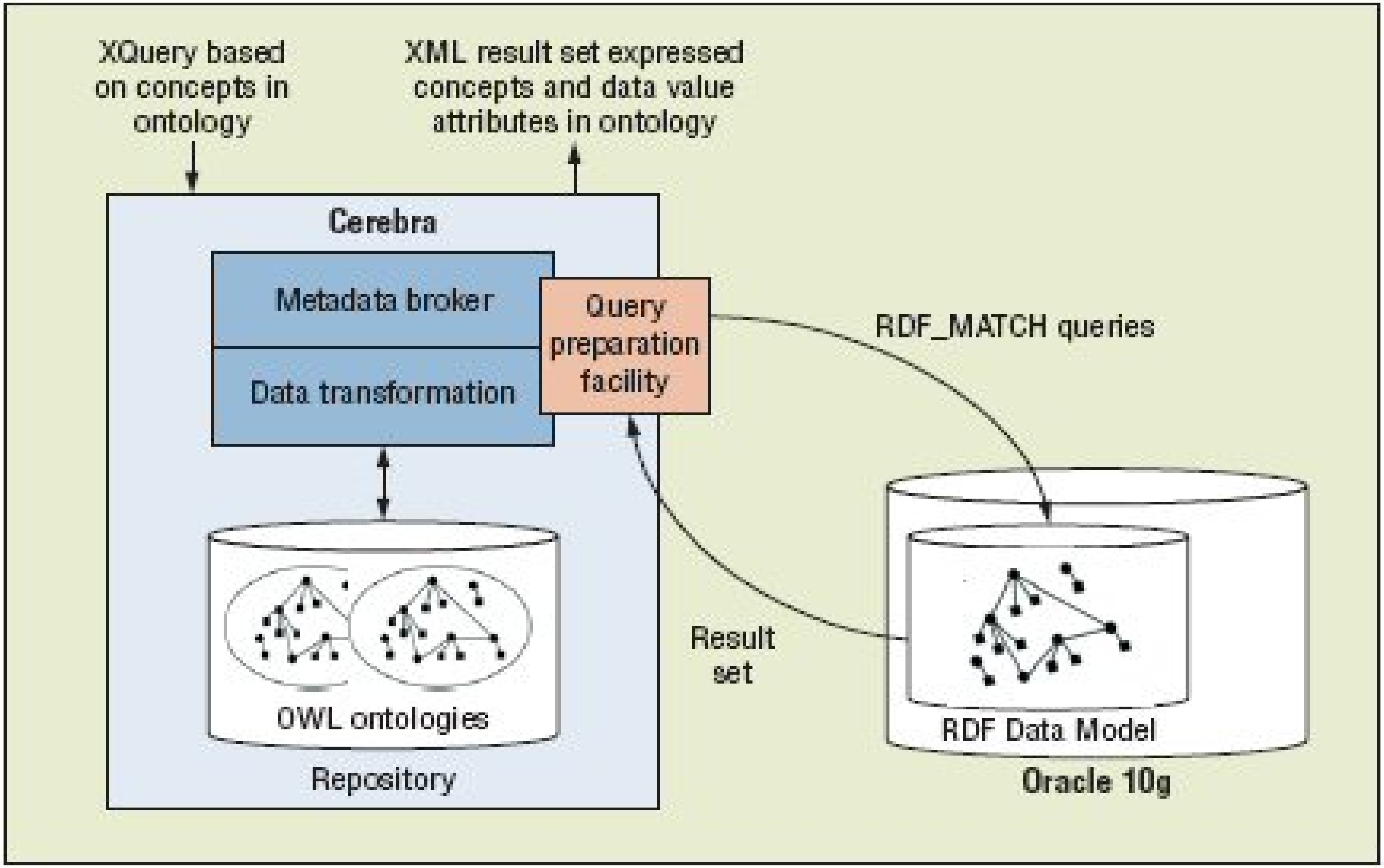





Image source: Drug Discovery Technology

# Ontology-based Data Integration





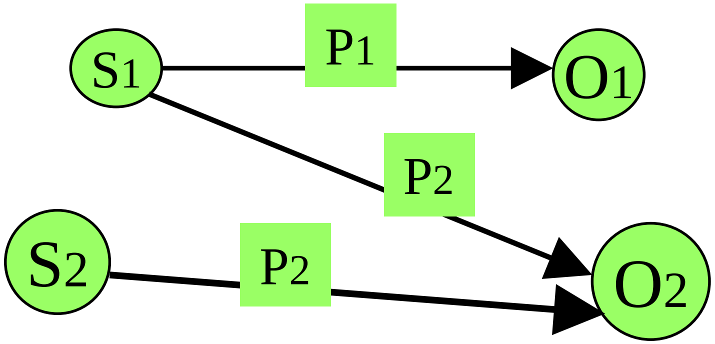

Source: IEEE Intelligent Systems

# Oracle and RDF: Motivation

- Customer requests
- RDF (and OWL) are maturing
- Oracle supports open standards
- Complements Oracle’s information management approaches
- Ability to leverage existing technologies

# Oracle RDF Data Model

- Support for RDF and RDFS
- Object-relational implementation
- Subjects and objects are re-used
- Links represent complete RDF triples




RDF Triples:

- {S1, P1, O1}
- {S1, P2, O2}
- {S2, P2, O2}

# SPARQL-like Query Capability

- A table function allows a graph query to be embedded in a SQL query
- Searches for an arbitrary pattern against the RDF data
- Includes inferencing based on RDF, RDFS, and user-defined rules
- Graph analytics available through Java API

# SDO_RDF_MATCH

```
SDO_RDF_MATCH (
     query   VARCHAR2,
     models   SDO_RDF_MODELS,
     rulebases SDO_RDF_RULEBASES,
     aliases   SDO_RDF_ALIASES,
     filter   VARCHAR2
)  RETURN  ANYDATASET;
```

# Enterprise Functionality

- Real Application Clusters (RAC), Security
- Multi-threaded, parallel processing, indexed, etc.
- Performance testing with UniProt

| | Q1 | Q2 | Q3 | Q4 | Q5 | Q6 |
|---|---|---|---|---|---|---|
| 10 M Triples | 0.86 | < 0.01 | < 0.01 | 0.03 | 0.18 | 0.46 |
| 20 M Triples | 0.95 | < 0.01 | < 0.01 | 0.03 | 0.19 | 0.47 |
| 40 M Triples | 0.96 | < 0.01 | < 0.01 | 0.03 | 0.18 | 0.47 |
| 80 M Triples | 1.03 | < 0.01 | < 0.01 | 0.03 | 0.20 | 0.49 |
| Maximum σ | .054 | 0.002 | 0.002 | .011 | .065 | 0.07 |

Units in seconds



Source: Chong et al. VLDB 2005

# Image Search

## “Find me all DICOM images that contain the term ‘Jaw’”

- Map relationships to terms using RDF triples

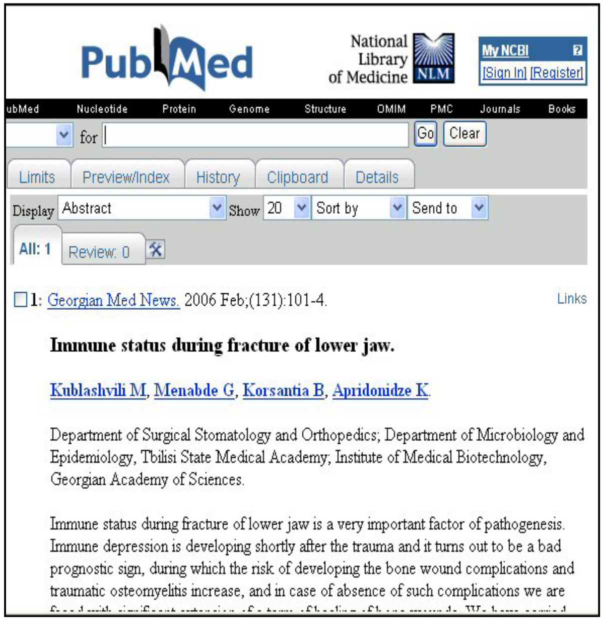
- ‘Mandible’, sameAs’, ‘Jaw’

- ‘Maxilla’, ‘partOf’, ‘Jaw’

# Text Search

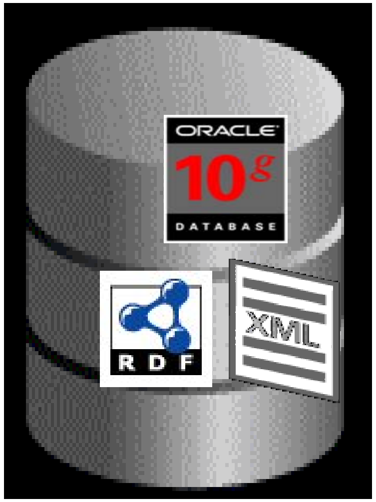
## “Find me all papers that contain the term ‘Jaw’”

- Map relationships to terms using RDF triples
- ‘Mandible’, sameAs’, ‘Jaw’
- ‘Maxilla’, ‘partOf’, ‘Jaw’



# Data Integration

- SQL / RDBMS
  - Concise, efficient transactions
  - Transaction metadata is embedded or implicit in the application or database schema
- XQuery / XML
  - Transaction across organizational boundaries
  - XML wraps the metadata about the transaction around the data
- SPARQL / RDF
  - Information sharing with ultimate flexibility
  - Enables semantics as well as syntax to be embedded in documents



# Integrating Biological Pathways

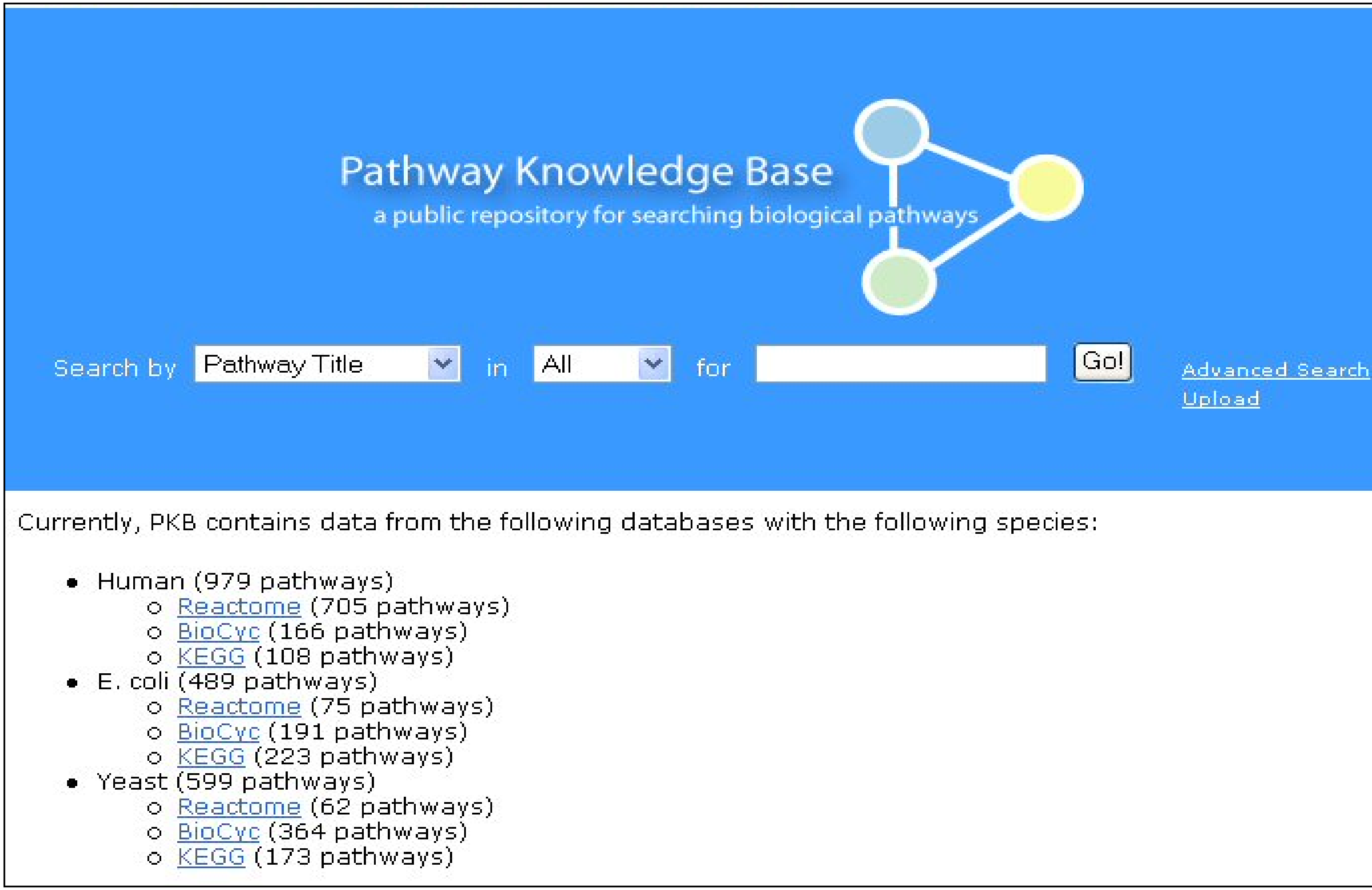





Image Source: Stanford University

# Integrating Data Across Discovery

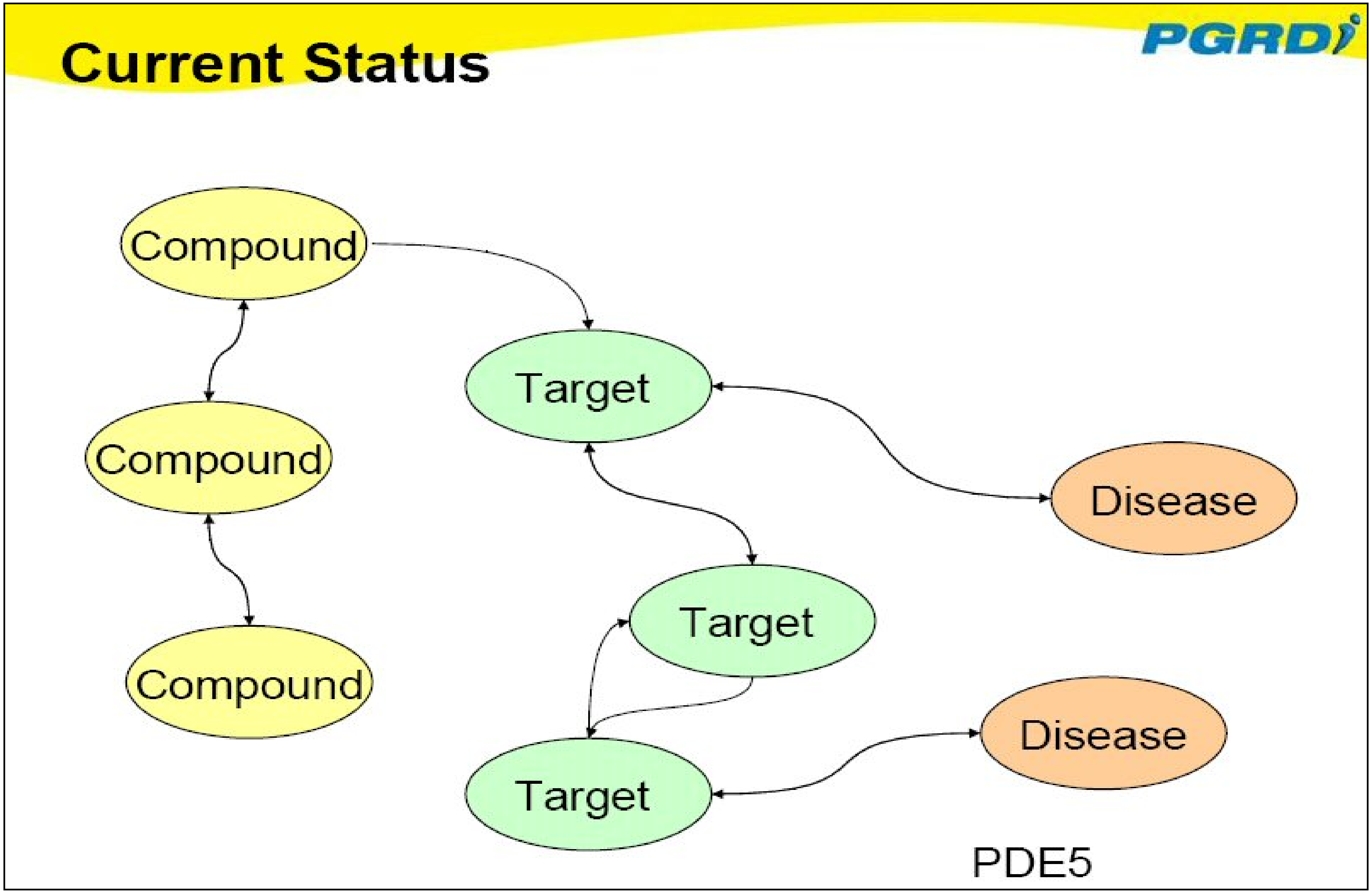
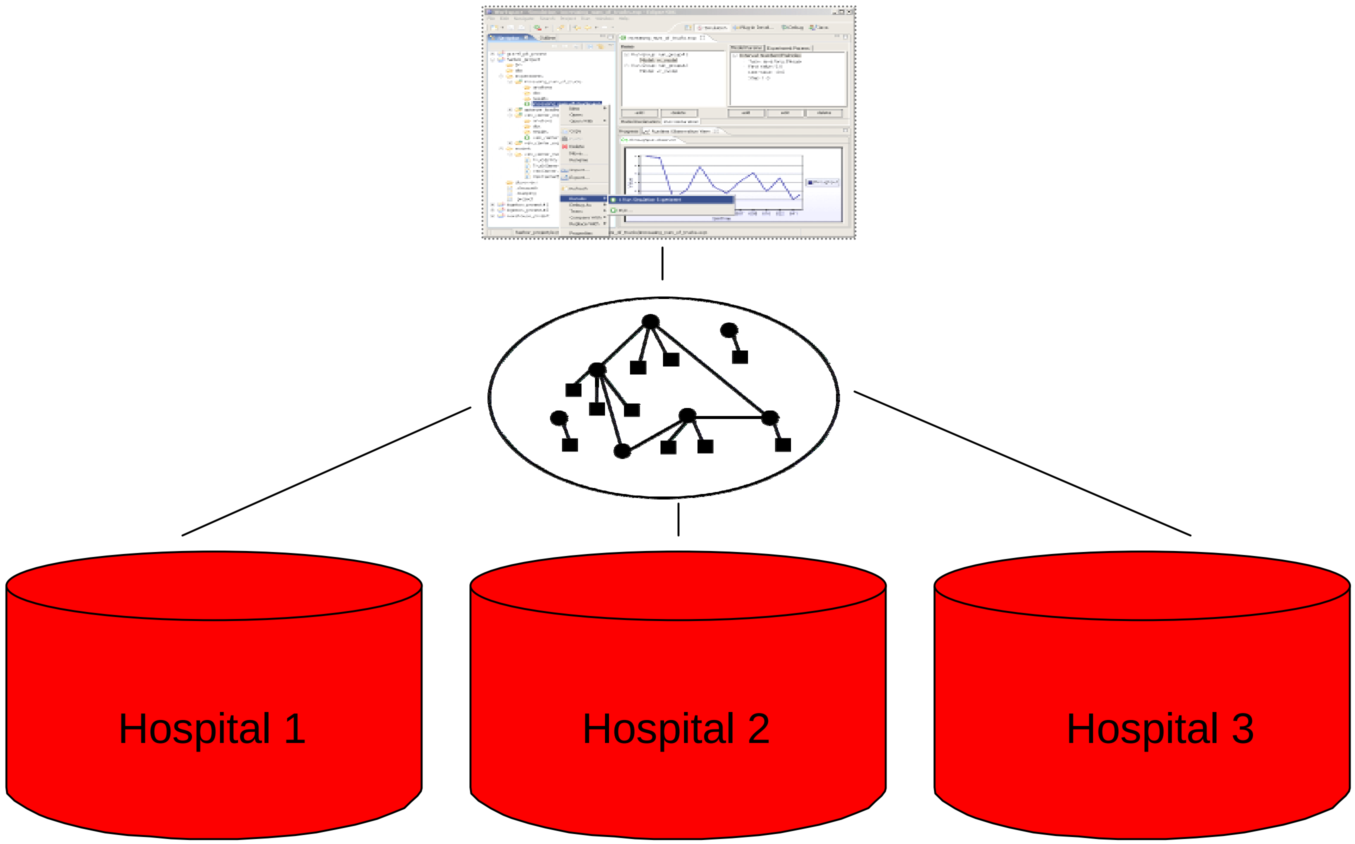





Image Source: Pfizer

# Cross Organization Data Integration



# Cross Domain Data Integration

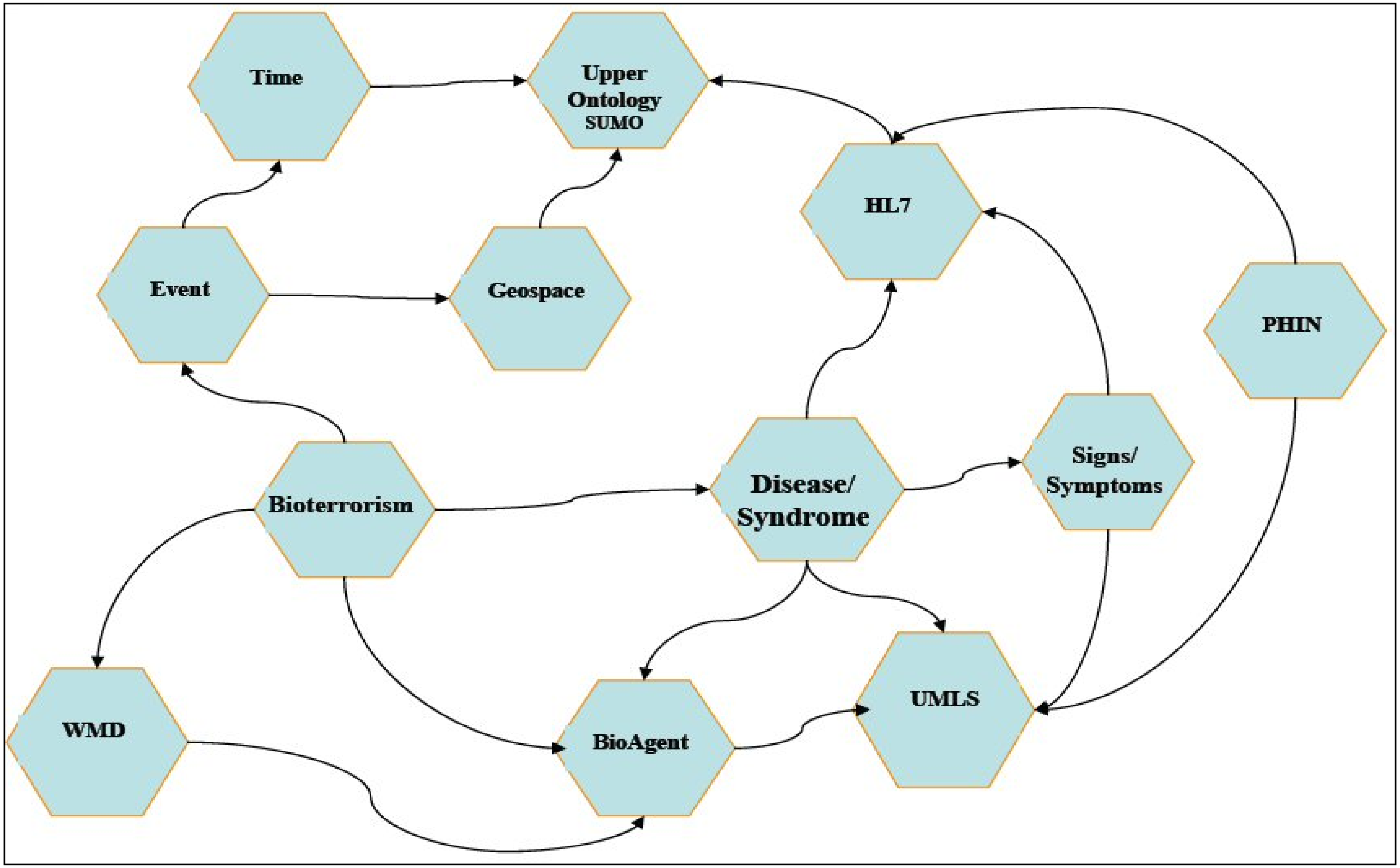





Image Source: University of Texas Health Science Center

# W3C Semantic Web for Health Care and Life Sciences Interest Group

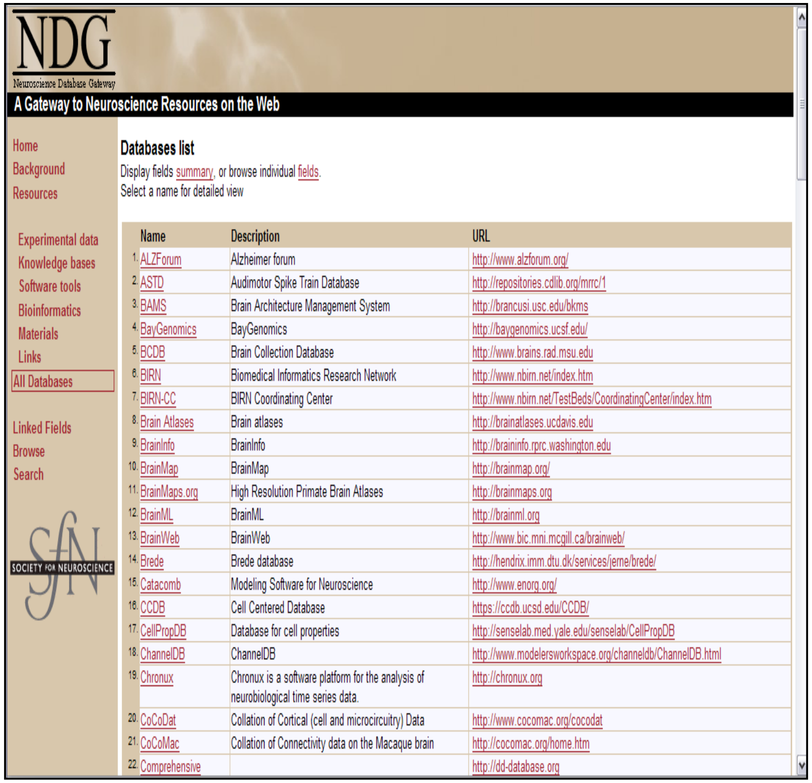
- Activities with HCLS
  - BioRDF
  - Ontologies
  - Adaptive Health Care Protocols and Pathways
  - Drug Safety and Efficacy
  - ROI Analysis
- Over 90 member organizations including Pfizer, Merck, AZ, AGFA, Partners Healthcare, Cleveland Clinic

# BioRDF: Seeding the Semantic Web

- Build a demo that spans bench to bedside to help scientist understand the value of the Semantic Web
- Explore the effectiveness of current tools for making data available as RDF
- Document finding to help accelerate adoption of the Semantic Web

# Neuroscience Focus

- Heterogeneous Data
  - Molecules to nervous system
  - Numerous Web resources
  - Effective data sharing and integration needed
- Disease Focus
  - Huntingtons
  - Alzheimers
  - Parkinsons



NDG
Neuroscience Database Gateway
A Gateway to Neuroscience Resources on the Web

Home
Background
Resources

Experimental data
Knowledge bases
Software tools
Bioinformatics
Materials
Links
All Databases

Linked Fields
Browse
Search

SOCIETY FOR NEUROSCIENCE

**Databases list**

Display fields summary, or browse individual fields.
Select a name for detailed view

| | Name | Description | URL |
|---|---|---|---|
| 1. | ALZForum | Alzheimer forum | http://www.alzforum.org/ |
| 2. | ASTD | Audimotor Spike Train Database | http://repositories.cdlib.org/mrrc/1 |
| 3. | BAMS | Brain Architecture Management System | http://brancusi.usc.edu/bkms |
| 4. | BayGenomics | BayGenomics | http://baygenomics.ucsf.edu/ |
| 5. | BCDB | Brain Collection Database | http://www.brains.rad.msu.edu |
| 6. | BIRN | Biomedical Informatics Research Network | http://www.nbirn.net/index.htm |
| 7. | BIRN-CC | BIRN Coordinating Center | http://www.nbirn.net/TestBeds/CoordinatingCenter/index.htm |
| 8. | Brain Atlases | Brain atlases | http://brainatlases.ucdavis.edu |
| 9. | BrainInfo | BrainInfo | http://braininfo.rprc.washington.edu |
| 10. | BrainMap | BrainMap | http://brainmap.org/ |
| 11. | BrainMaps.org | High Resolution Primate Brain Atlases | http://brainmaps.org |
| 12. | BrainML | BrainML | http://brainml.org |
| 13. | BrainWeb | BrainWeb | http://www.bic.mni.mcgill.ca/brainweb/ |
| 14. | Brede | Brede database | http://hendrix.imm.dtu.dk/services/jerne/brede/ |
| 15. | Catacomb | Modeling Software for Neuroscience | http://www.enorg.org/ |
| 16. | CCDB | Cell Centered Database | https://ccdb.ucsd.edu/CCDB/ |
| 17. | CellPropDB | Database for cell properties | http://senselab.med.yale.edu/senselab/CellPropDB |
| 18. | ChannelDB | ChannelDB | http://www.modelersworkspace.org/channeldb/ChannelDB.html |
| 19. | Chronux | Chronux is a software platform for the analysis of neurobiological time series data. | http://chronux.org |
| 20. | CoCoDat | Collation of Cortical (cell and microcircuitry) Data | http://www.cocomac.org/cocodat |
| 21. | CoCoMac | Collation of Connectivity data on the Macaque brain | http://cocomac.org/home.htm |
| 22. | Comprehensive | | http://dd-database.org |

# BioRDF Tasks

ESW Wiki Login

HCLSIG BioRDF Subgroup | Tasks

FrontPage | RecentChanges | FindPage | HelpContents | **Tasks**

Edit (Text) Edit (GUI) Info Attachments More Actions:

**Active Tasks**

- /Reagents (Status)
- /SenseLab
- /Using SW Technologies to Find Small Molecules that Bind to Proteins
- /Entrez Gene to RDF
- /OMIM Neural diseases
- /Natural Language Processing and RDF
- /Ligand-Receptor Interaction, Molecular Interaction Networks, Ontology Evolution
- /Vocabulary Requirements
- /URI Best Practices

**Proposed Tasks**

- /Brain Connectivity
- /Brain Atlas condition to scans
- /Protein Neural related protein data
- /Ruby On Rails and ActiveRDF



Source: http://esw.w3.org/topic/HCLSIG_BioRDF_Subgroup/Tasks

# Lessons Learnt

- It’s a hybrid world
  - Relational, XML & RDF, OWL
  - Triple stores and on the fly mappings
- Many applications of the Semantic Web
- Typically minimal overhead to getting started
- Use existing identifiers as URIs
- Oracle’s original RDF loader was slow
- Support for OWL required

# Summary

- The Semantic Web offers heterogeneous data integration using explicit semantics
- Oracle has a scalable, secure, highly-available RDF Data Model
- Many organizations are adopting semantic technologies
- W3C is actively involved in helping the life sciences and health care embrace the Semantic Web